

Physiology of the digestive system



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The oral cavity

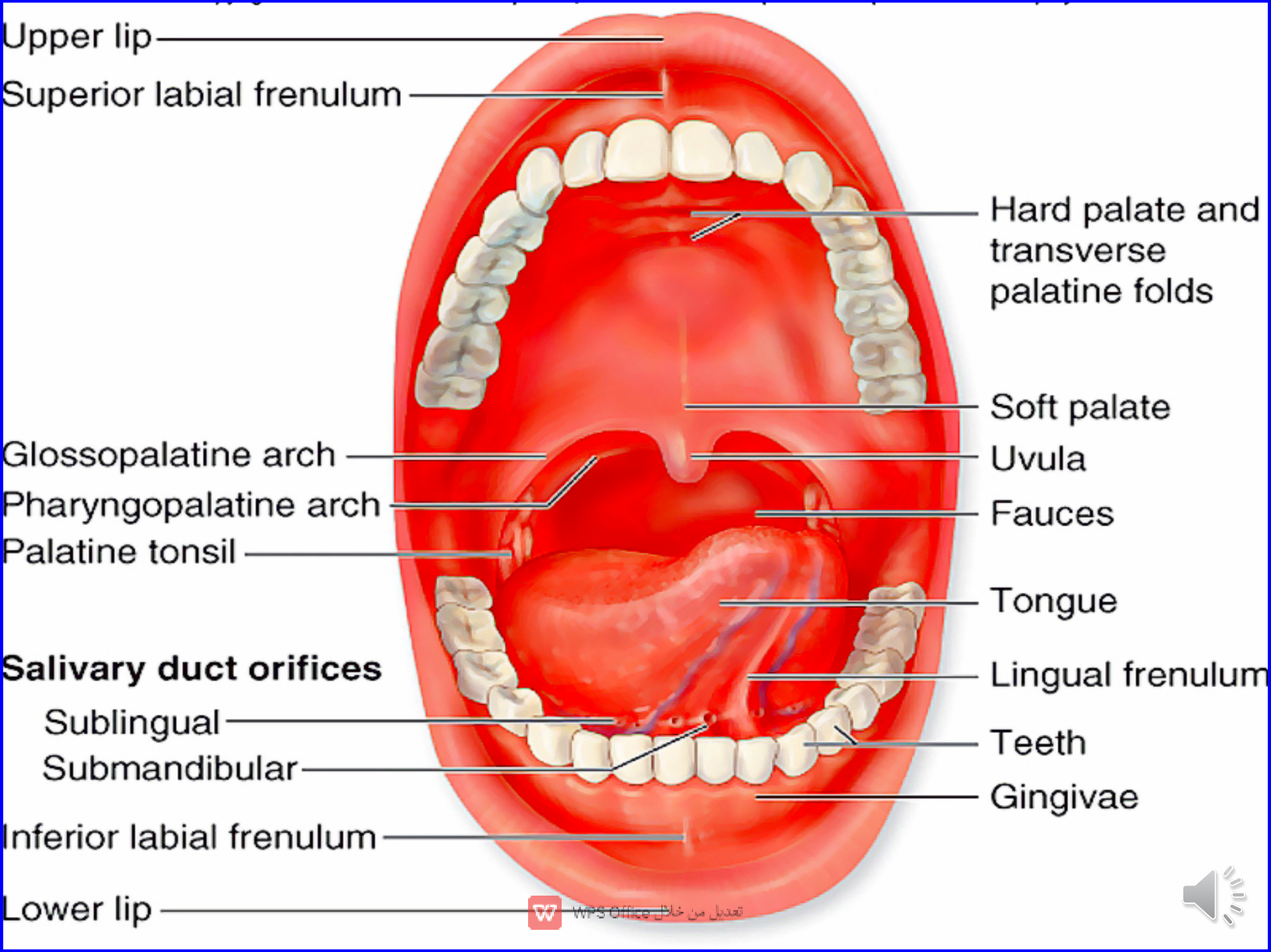
➤ Functions:-

- Mastication (the mechanical breakdown of food).
- Mixes food with saliva.
- Initiates swallowing.

➤ Saliva:-

- Fluid poured into the oral cavity produced by salivary glands.





Mastication (chewing)

- Is the mechanical breakdown of solid particles into smaller ones that would be absorbed more easily
 - Muscles of mastication
 - Palate
 - Tongue
 - Cheeks
 - Teeth



Muscles of Mastication

- Mechanism:-
- Has a voluntary & reflex phases
- Muscles of mastication are :- masseter , medial pterygoid , temporalis, lateral pterygoid & digastric
- Pressure of food against receptors in the mouth causes relaxation of jaw closing muscle & contraction of jaw opening muscle so the jaw opens
- Loss of stimulus & stretch of the jaw causes the jaw to close



- The anterior teeth – a cutting action
- The posterior teeth – a grinding action
- The maximal **closing force** – incisors 15 kg
- – molars 50 kg
- **Innervations** of the muscles of chewing – 5th, 8th, 12th cranial nerves
- **Centres** – near the brain stem and cerebral cortex centres for taste



Muscles of Mastication

- Masseter---closes
- Temporalis---closes
- Medial pterygoid---closes
- Lateral pterygoid----opens
- Innervation by 5th cranial nerve (trigeminal), mandibular branch



Tongue

- Houses Taste receptors
- Speech formation
- Mastication
- Swallowing
- Chewing also stimulates the saliva secretion

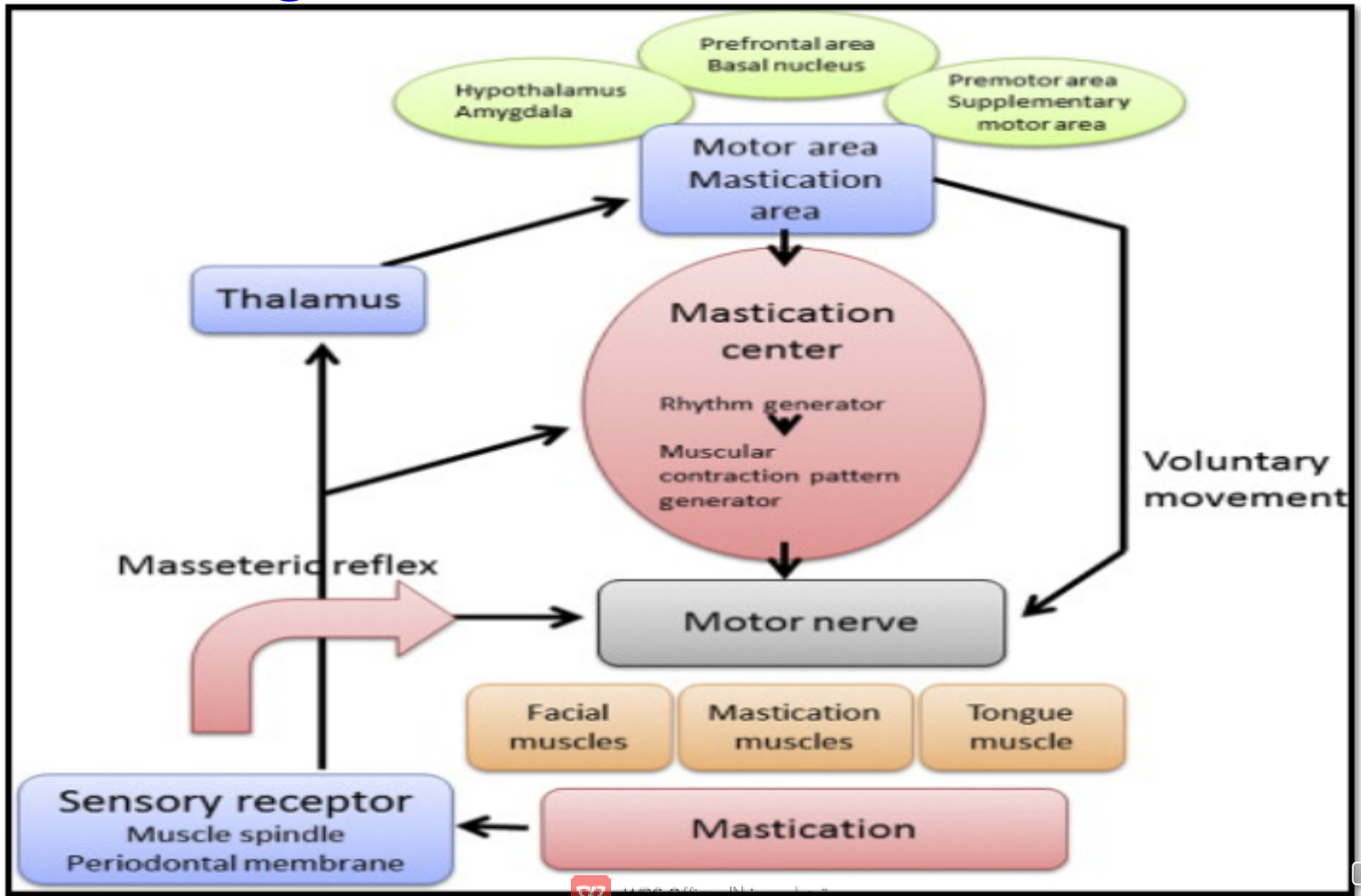


Regulation of mastication

- It is an automatic, rhythmic process like walking and breathing
- Can be modified (faster or slower)
- Centre is located in the lower brain stem
- Higher brain can also control it (cortex, amygdala, RF, hypothalamus...)



Regulation of mastication



Mastication disorders

- **Masticatory muscles disorder:**
 - Myalgia (muscle pain)
 - Spasm
 - Jaw movement disorder (TMJ)
 - Myositis



End of the lecture

